

Finding the Sign Errors in Parabola Transformations

- In $y = a(x - h)^2 + k$ the vertex is (h, k) . The trap is the bracket: $x - 3$ moves the graph 3 units **right**, and $x + 3$ moves it 3 units **left** — the opposite of how each one reads.
 - The sign of a decides the direction: $a < 0$ flips the parabola over and makes the vertex a maximum. The sign of k raises or lowers it.
 - Where a student's work is shown, it contains **exactly one** error. Find it before you compute anything.
 - A single substitution checks a transformation: put one x -value into the equation and see whether the point you get matches the graph you expect.
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1. The graph of $y = (x - 5)^2$ touches the x -axis at exactly one point. Solve $(x - 5)^2 = 0$ to find the x -coordinate of that point.

Answer: _____

2. Find the x -intercepts of $y = (x + 4)^2 - 1$ by solving $y = 0$.

Answer: _____

3. For $y = -2(x - 3)^2 + 5$, find the value of y when $x = 1$.

Answer: _____

4. Mia is asked for the value of $y = -(x + 2)^2 + 7$ when $x = 1$. She writes:

$$y = (1 + 2)^2 + 7 = 9 + 7 = 16$$

- (a) Name the sign error in Mia's work. Say which of her steps is the wrong one.
(b) Compute the correct value of y when $x = 1$, and put it on the answer line.

Answer: _____

5. Find the x -intercepts of $y = -x^2 + 6x - 5$ by solving $y = 0$.

Answer: _____

6. Owen is asked for the x -intercepts of $y = 2(x + 3)^2 - 8$. He writes:

$$2(x + 3)^2 = 8, \text{ so } (x + 3)^2 = 4, \text{ so } x + 3 = \pm 2, \text{ so } x = 5 \text{ and } x = 1.$$

(a) Name the sign error in Owen's last step. Everything before it is correct, so be precise about which step fails.

(b) Give the correct x -intercepts on the answer line.

Answer: _____

7. The parent graph $y = x^2$ is reflected in the x -axis and then shifted 4 units *down*.

(a) Write the equation of the new graph.

(b) Evaluate the new function at $x = 3$, and put that value on the answer line.

Answer: _____

8. Synthesis challenge. $y = -3(x + 2)^2 + 12$

- (a) Write the vertex, and say whether the parabola opens upward or downward.
- (b) Find the x -intercepts, and put them on the answer line.
- (c) Explain what the sign of each of the three numbers -3 , $+2$ and $+12$ does to the graph, and say which of those signs students most often reverse.

Answer: _____